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HIPAA and Dental Information Systems

Abstract

Electronic dental records (EDRs) and electronic health records (EHRs) are two major areas of progress in modern medical practice. These advancements give patients a lot of important information about their health. This information is private, so it's really important that all healthcare providers follow the rules set by HIPAA.. This research paper focuses on the relationship between HIPAA regulations and dental information systems. It evaluates how the HIPAA Privacy Rule, Security Rule, and Breach Notification Rule impact the overall architecture of the dental information system as well as data protection practices and interoperability in healthcare IT systems for dentistry. The paper explores current cybersecurity threats to dental information systems, including ransomware attacks, unofficial access attempts, and issues related to misconfigured systems. Given the complexities associated with oral and dental health information, specifically with the management of diagnostic imaging, pathology reports, and other types of information that require additional handling and protection protocols; this data faces unique challenges. Research indicates that adhering to HIPAA standards and regulations in dental information systems is largely reliant upon design and security of secure system as well as implementation of strong encryption, strict controls over access to the system, and complete audit records (NIST, 2024). In conclusion, complying with HIPAA guidelines will allow for the proper use of patient confidentiality in providing dental health care, quality of data contained in dental health information systems, and the secure digital transformation of dental health care.

Introduction

The

shift from traditional papervrecords to digital formats used to manage clinical, administrative and

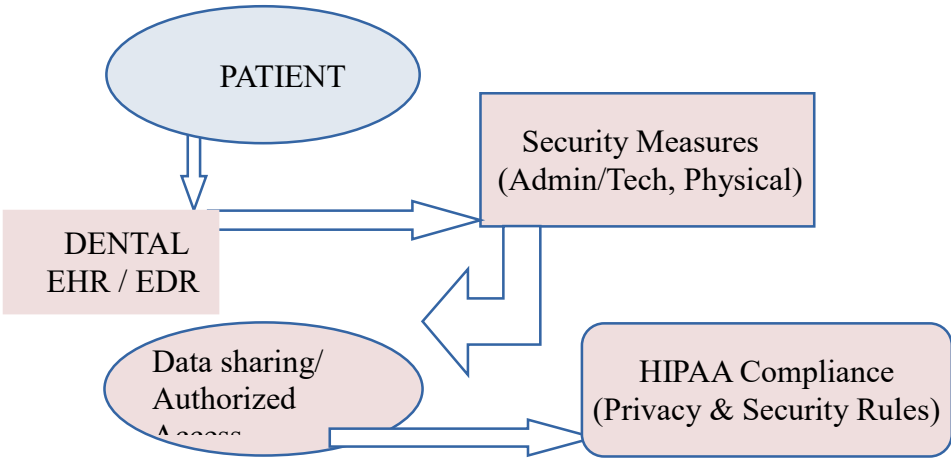
financial information about patients (Scarano et al., 2025) provides many benefits; including operational efficiencies and improvements in the service quality provided to patients. The increasing digitization of health-related data creates a growing number of legal and regulatory issues for protecting the privacy, security, and proper use of patient data (TDIC Risk Management Staff, 2023). The Health Insurance Portability and Accountability Act (HIPAA), Enacted by the U.S. Department of Health and Human Services in 1996, sets forth various rules to protect the privacy of sensitive health information (Edemekong, 2024) as well as provide individuals with their health information. Administrative Simplification may be relevant to a dental practice information system, including provisions designed to create a unified framework of regulatory requirements for electronic health information (ePHI) (Chasteen et al., 2003). The Administrative Simplifications contain many significant mandates concerning the protection of electronic health data including: the Privacy Rule, the Security Rule, and the Electronic Transactions and Code Sets Rule. Kevin Henry (2025) noted that According to Kevin Henry (2025), for an organization to be compliant with HIPAA, it is important that they have technical safeguards in place to ensure strict compliance with the rules, such as encryption, access control, audit mechanisms, and legal knowledge. The primary reason for these technical safeguards is to reduce the risk of data breaches and to create a safe and secure environment where users can maintain their trust in the digital health space. This paper states that although HIPAA is an important legal mechanism for protecting dental health data, the flexibility of the regulation (as well as lack of specificity) provides significant gaps for small dental practices in relation to being compliant with their cybersecurity preparedness, interoperability and utilization of new technology.

Legal Framework: HIPAA and Dental Providers

HIPAA classifies dental office as covered entities under the privacy rule when they electronically communicate any health information for specified transactions (U.S. Department of Health & Human Services, n.d.). The Health Insurance Portability and Accountability Act of 1996 (42 U.S.C. § 1320d et seq.) provides the statutory foundation for protecting health information, including dental records (U.S.

Department of Health & Human Services, n.d.). The HIPAA Privacy Rule (45 C.F.R. Part 164, Subparts A & E) sets forth specific guidelines on how covered entities may use and disclose an individual's identifiable health information. The Security Rule (45 C.F.R. Part 164, Subpart C) requires administrative, physical, and technical safeguards to protect electronic PHI (ePHI) (U.S. Department of Health & Human Services, 2002). These requirements also apply to dental electronic health records, including images and pathology reports (Schleyer et al., 2011). Specifically, 45 C.F.R. 164.502 and 164.524 were developed to govern access to confidential information about patients (U.S. Department of Health & Human Services). HIPAA also imposes two requirements: compliance with technical, physical, and administrative safeguards; and the right to an accessible electronic health-records system (Schwendicke et al., 2025). HIPAA does not stipulate certain technologies that dentists should use; thus it requires dentists to individually take these regulations into account and determine compliance (Simon et al., 2019). This could lead to varying levels of compliance within the dental community, particularly among smaller dental practices that have limited experience with cybersecurity (Simon et al., 2019). The HITECH Act (2009) also contributed to the enforcement of regulations relating to access to and Transfer of Health Information by requiring all health care providers using third-party vendors to comply with encryption and privacy standards set forth by the federal government (U.S. Department of Health & Human Services, 2009; 2013). In order to provide compliance, dental practitioners need to develop and enforce their own policies regarding access to patients' information, protecting its integrity, and notifying patients when there has been a breach, consistent with the two-pronged approach taken by HIPAA regarding the privacy and security of protected health information (U.S. Department of Health & Human Services, 2011). However, when dental practitioners have to deal with many operational obstacles in the process of providing patients with access to their records, such as accessing large digital image files including digital radiographs, such as through electronic storage devices or transmission systems, or accessing large digital image files (including digital radiographs) via a networked connection to a central

database, they may encounter significant operational difficulties in their ability to provide their patients with access to their records (Schleyer & Mendez, 2011).



Also, there's a lot of new technology coming out like tele-dentistry, AI, cloud computing, etc., that are becoming common place. But currently HIPAA addresses these issues within their guidelines, but very few provide specific framework for implementation and execution, so dental practices are having the hardest time achieving this consistently. In June 2022, Mayfield Dental Arts in Ohio reported a breach where a former employee accessed 3,318 individual patient records with sensitive PHI and accessed all (DOB, SSN, Lab results, etc.). This case demonstrates some of the practical aspects of HIPAA compliance (generally), especially with respect to the need for good administrative and incident response plans for small dental practices (Simon et al., 2019). Additionally, this example shows how difficult it can be to detect insider threats (and these are still a very significant area of vulnerability with HIPAA).

Table 1: HIPAA Requirements, Applications, & Implementation Challenges in Dental Health IT

HIPAA Rule	Purpose	Application in Dental Systems	Key Challenges	Case Example
Privacy Rule (45 C.F.R. §164.502)	Secures privacy of PHI	Patient records, treatment plans	Managing access across staff roles	Mayfield Dental Arts, OH (2024): Former employee accessed 3,318 patient records. HIPAA notifications issued; Highlights strict access controls & staff training.

Security Rule / Safeguards (45 C.F.R. §164.306–318)	Safeguards ePHI technical, physical, administrative	Encryption, access controls, secure EDR systems	Limited IT resources in small practices	Jefferson Dental Center, IN (2024): Ransomware affected 12,340 patients; safeguards strengthened. Professional Dental Alliance, FL (2019): Employee improperly accessed ePHI; penalties issued & staff retrained.
Patient Access (45 CFR §164.524)	Grant patient access to their records	Access to Digital radiographs and reports	Large file sizes & system restrictions	Cambridge Dental Consulting Group, NV (2018): Business associate exposed ePHI of 3,758 patients online; OCR guidance issued & safeguards improved. Highlights robust HIPAA access, audit, & compliance
Breach Notification (45 C.F.R. §164.400–414))	Requires reporting of data breaches	Reporting ransomware or unofficial access	60 day reporting window & risk assessment	Well Care Health Plans, Inc., FL (2017): Business partner mis-mailed 1,214 member records; notifications made & corrective measures implemented to prevent recurrence.
Enforcement Rule (45 C.F.R. §160.300–316)	Establishes penalties and procedures for HIPAA violations	Ensures dental office & vendor abide with HIPAA through audit, inquiry & penalties	Limited compliance awareness in small practices; financial burden of penalties	Family Dental Care, P.C., IL (2020): Failed to provide a patient’s complete records within the required time frame (over 5 months delay), violating HIPAA Right of Access. Settled with a \$30,000 penalty and implemented corrective actions.

(Source: P. F. Edemekong et al.,2024; Schleyer et al.,2011; Simon et al.,2019, HHS Civil rights office)

In August 2024, P.D. Park Dental (MN) reported a phishing scheme affecting 185,606 people. This practice improved access controls, provided training for staff and informed the Department of Health, the affected individuals and the media. OCR (Department of Health and Human Services) provides technical assistance. This highlights the continuing challenge of implementing strong administrative and technical guarantees in dental practices (Schwendicke et al., 2025; Simon et al., 2019).

Dental and Oral Health Information system

Electronic Dental Records (EDRs) include clinical notes, treatment plans, radiographs, and pathology reports. In comparison with general medical EHRs, dental systems incorporate special imaging method , intraoral & panoramic X-rays, and connect with dental laboratories to perform analysis of

prosthetics, orthodontic and pathology (Schleyer et al., 2011). Dental EHRs are smaller in scope and complex in workflow compared to hospital EHRs (Schwendicke et al., 2025). They tend to work at smaller practice units and have fewer IT resources that can affect the robustness and security of the system (Simon et al., 2019). The customization of dental information requires specific software solutions and interoperability standards, which make it difficult to share data with the rest of the health information network (Schwendicke et al., 2025). Dental information systems need to deal with structured data like treatment codes, periodontal charting, and unstructured data (Schleyer et al., 2011). Numerous imaging systems are based on a standard e.g. DICOM (Schwendicke et al., 2025). These standards are effective in maintaining consistent image formatting but are not fully integrated with the larger EHR systems, which hamper interoperability (Schwendicke et al., 2025). Interoperability limitations can impede HIPAA compliance by limiting efficient data exchange, mostly for comprehensive patient data, which reduces both clinical effectiveness and legal compliance.

Table 2: Dental EHR vs Medical EHR

Feature	Dental EHR	Medical EHR
Data Type	Imaging heavy (X-rays)	Text & lab results
Standards	DICOM (limited integration)	HL7, FHIR
Practice Size	Small clinics	Large hospitals
Interoperability	Limited	More advanced
IT Resources	Often limited	More robust

(Source: Scarano et al., 2025; Simon et al., 2019)

Dental Information Technology (DIS) Challenges

The IT systems used to deliver dental health services have vulnerability to large data security threats. Ransomware, unauthorized access, and insider threats are some of the data security threats that the IT systems provide support to the dental healthcare systems (Fernandez-Aleman, et al., 2013; Schwendicke, et al., 2025). Because dental practices often have limited IT budgets and are smaller in size, many of them have inadequate cybersecurity systems (Gupta, et al., 2025; Simon, et al., 2019).

DIS systems are especially vulnerable to ransomware attacks, phishing, insider threats, and unwanted remote access, as well as many other types of cybersecurity risks (Zakaria S, 2024). Interoperability is hampered by the lack of standardized format and protocols for exchanging dental health information between providers (Schleyer, et al., 2011; Schwendicke, et al., 2025)(Schleyer, et al., 2011; Schwendicke, et al., 2025). Access controls for role-based access control (RBAC) need to be implemented to ensure that only those staff members who have been granted access can view sensitive dental data (Zakaria S, 2024). As stated by the U.S. Department of Health & Human Services (n.d.), there is also increased risk associated with vendors and other business associates that utilize a third party software vendor or contractors to manage ePHI. Consequently, vendors must be evaluated to ensure compliance with HIPAA Security Standards and Requirements, particularly among smaller dental providers (Simon et al., 2019). Due to a lack of knowledge on interpreting HIPAA standards, small providers may create varying degrees of data security as well as differential levels of protection for patients data. Consequently, ransomware attacks pose a significant threat to dental providers. The presence and availability of patient records is critical to the functioning of a dental office, making the loss of those records completely debilitating to the operation of a dental practice (Jack et al., 2024). Therefore, if providers cannot recover their data in a timely manner after it has been encrypted, they may experience severe disruption in their practice and may face increased penalties for violating HIPAA (U.S. Department of Health & Human Services, n.d.). Security breaches often result from simple human errors, so routine education of employees on achieving the best possible balance between usability of and security of the computer systems is critical. A proper balance between usability and security is always going to be a priority. Some examples are: providing multiple means of verifying an employee's identity (multi-factor authentication; beyond just entering a password) and implementing strict access control methods to comply with HIPAA regulations. While secured systems support good workflow habits, a lack of workflow efficiency can lead to employees violating established security procedures, which create much greater potential for vulnerabilities in those systems.

Table 3: Cybersecurity and Policy Challenges in Dental Health IT

Challenge	Description	Impact On Dental Practices	Possible Mitigation
Ransomware	Data is encrypted by attackers	Operational disruption, HIPAA violation	Regular backups, endpoint security
Data Silos	Medical dental EHRs not integrated	Limits continuity of care, repeated tests	Interoperability standard -DICOM, HL7
Tele-dentistry	Remote dental care with patient images	Regulatory ambiguity, cross jurisdiction issues	HIPAA compliant tele-health platforms, BAAs
Cloud Storage	Third party data hosting	Cross border storage risks, unclear liability	Enforce BAAs, encrypt data at rest

(Source: Simon et al., 2019, Jack 2024, Gupta et al., 2025, Schleyer et al., 2011)

Health Policy and Regulatory Challenges

Due to their flexibility and lack of standardized requirements, HIPAA's regulatory model is subject to varied degrees of dental practice security based on varying amounts of IT knowledge of the practice (U.S. Department of Health & Human Services, n.d.; Simon et al., 2019). Other similarly situated organizations can implement varying levels of protection depending on their circumstances, disrupting the consistency in protecting patient records (Schwendicke et al., 2025). The trends in enforcement suggest that fewer dental-related compliance actions are taken than other parts of the health care industry (U.S. Department of Health & Human Services, n.d.), possibly due to the underreporting of violations or leniency toward violators. Cases like Mayfield Dental Arts highlight the need for better guidance related to access controls in the dental practice, as well as training for staff on dental practice security (U.S. Department of Health & Human Services, n.d.). It may also be necessary to update HIPAA regulations so that they provide additional clarity regarding the management of insider threat risks for small physician IT systems. There is currently a disconnect between federal HIPAA guidelines and many state dental privacy laws, making compliance difficult within this complicated patchwork of regulatory frameworks (U.S. Department of Health & Human Services). As such, it can be difficult to

manage risk within any environment, and there may be unequal levels of protection offered to patients depending upon their geographic location. For example, an email phishing attack at PDG, P.A. dba Park Dental (MN, 08/09/2024) impacted the protected health information (PHI) of approximately 185,606 affected individuals because of the targeting of multiple employees. The enhanced access control measures and workforce security awareness training were implemented via the policies established by the Covered Entity (CE) to ensure increased workforce security through enhanced access control policies (U.S. Department of Health & Human Services, date not available). The CE will provide notification to the Department of Health and Human Services (HHS), the affected individuals, and the media (U.S. Department of Health & Human Services, date not available) that any complaint made against a Covered Entity (CE), in relation to an OCR complaint, will be handled appropriately with technical assistance from the Office for Civil Rights (OCR) regarding HIPAA compliance. There are many reasons to highlight the importance of having an established set of administrative and technical assurances for dental services (Schwendikke et al., 2025). There are questions about the ways in which dental services comply with regulations regarding the confidentiality and privacy of patient and staff data. It has become increasingly difficult to determine manufacturers legal obligations in terms of secure electronic transmission of dental imaging files and the legality of the provision of dental services across borders (Estai et al., 2018). Telehealth is subject to the Health Insurance Portability and Accountability Act (HIPAA), although only general guidelines exist and few guidelines that specifically apply to the practice of tele-dentistry have been published. Dental practitioners using tele-dentistry or any type of remote dental technology may be uncertain about how to proceed with providing their patients with care. The use of electronic health records has further complicated compliance with respect to multi-jurisdictional storage of data (Fernandez-Aleman et al., 2013). More than 3 million patients protected health information (PHI) has been compromised as a result of breaches or hacking within the health care industry (U.S. Department of Health & Human Services, n.d.). Given the high occurrence of data breaches within the healthcare field that result from human error, the dental industry must train its

employees to comply with HIPAA (Simon, et al., 2019). Scenario-based training (e.g., phishing simulation, role-based training and gamification) must be continuous to obtain a reasonable level of compliance with HIPAA regulations. IT consultants or managed service providers provide support to dentists as they implement encryption, set up access controls and monitor their information systems (NIST, 2024). The use of these providers can enhance an organization's cyber governance, aid in correcting technical deficiencies and help maintain laws. Cybersecurity threats from both ransomware and insider misuse remain significant challenges, as they can directly affect patient confidentiality and access to your systems. Additionally, with recent changes to regulations for HIPAA Notice of Privacy Practices (NPP) being introduced to further protect SUD treatment records that begin on 16 February 2026, all covered entities, such as dental practices and other health care providers will be required to amend their NPPs to comply with the new legislation or risk being non-compliant. The following five NPP amendments must be made by the 16 February 2026 deadline in order for you to maintain compliance. The first of the five amendments requires that when providing access or releasing any SUD treatment records, written consent must be obtained from each patient prior to doing so. The second change involves further protective measures being provided for SUD data requiring formalization. The third change allows a patient to withdraw their written consent at any time after giving it. This allows individuals to take more control over who has access to their SUD treatment record. The fourth change also includes advising each patient of the risk associated with their SUD record being published or re-published after being released to someone else for treatment as part of the NPP. Finally, the fifth change strengthens patient rights to opt out of receiving any communications relating to fundraising activities. To ensure that patients receive clear information and transparency on the updated NPP documents, dental practices should have the latest version of the NPP publicly available for all patients, at a minimum through posting a link to the NPP document on the dental practice's website (US HHS OCR, 2024).

Every dental organization must create a comprehensive risk management plan that covers the entire scope of its IT domain, including conducting regular security assessments of its IT systems and providing training for employees (Schwendicke et al., 2025). The administrative burden placed on organizations by HIPAA creates both financial and operational issues for all organizations, and, particularly for small businesses with limited resources, this burden can be extremely challenging to absorb (Rittenhouse et al., 2017). The establishment of effective governance structures, through the creation of clearly defined responsibilities, is vitally important to developing an organization's legal obligations and, therefore, reducing an organization's exposure to potential liability. Lastly, it is also important for dental practices to require third-party vendors to enter into Business Associate Agreements (BAA) with them as required by HIPAA (U.S. Department of Health & Human Services, n.d.). Conversely, small dental practices have limited amounts of legal assistance available for the purpose of reviewing those agreements thoroughly (Simon et al., 202019). For this reason, many small dental practices struggle to provide basic protection from the rapid advancement of possible threats. The potential risks to patient privacy and security related to protected health information (PHI) have long been of concern, and the use of artificial intelligence (AI) to access and process large amounts of patient image and clinical data and other protected health information will only increase those privacy and security concerns and risks (Jayatissa, et al. 2023). The ability to have interoperability between AI algorithms and dental electronic medical record (EMR) systems should allow for secure, auditable, and consent based exchange of data from one another. In this case, security can be defined as encryption, access control (NIST, 2024), and tracking of how algorithm driven data is used. The correct implementation of AI in clinical practice has the potential to improve both the way that diagnoses are formed and the way that decisions are made within the clinical setting; however, due to a lack of adequate safeguards to ensure compliance with HIPAA, the practitioners may be at risk for violation of regulatory or legal requirements (Schleyer, et al. 2011). Due to the increased utilization of technology like cloud based dentistry and tele-dentistry, practitioners now have the capability to conduct consultations long distance, send images safely and easily to specialists for review

and collaborate easily with laboratories, (Estai et al. 2018). However, the availability of these new technologies poses a challenge regarding the storage and transmission of data across multiple jurisdictions, requiring significantly greater effort to ensure compliance with HIPAA's Privacy and Security Regulations (Henry, 2025). To mitigate risk associated with an organization's use of either a distributed infrastructure or a third-party vendor will continue to require Business Associates Agreements, encryption, and role-based access controls. Therefore, finding a balance between proper legal compliance and the growing technological basis upon which the dental industry operates will be critical, (Rittenhouse et al., 2017). The following are necessary to ensure compliance with HIPAA by using AI, Cloud Computing, and Tele-Dentistry to improve Quality of Life: ongoing monitoring of IT systems; training and education of staff members; and a proactive approach to managing risk in order to achieve Compliance with HIPAA (Zakaria, 2024 & Estai et al., 2018). These technologies are where Innovation occurs and allows Patients to Trust Providers with their Electronic Health Information. As Cloud Storage and Tele-Dentistry Services become available, Providers must ensure compliance with HIPAA by implementing Security Controls, such as encrypting information at rest and in transit as well as using secure portals and multiple factor authentication.

Conclusion

The

Health Insurance Portability and Accountability Act legally protect dental and oral health records (U. S. Department of Health & Human Services, n.d.). However, there are still challenges when trying to implement these laws within the dental information systems of technology. The challenges to implement the HIPAA regulations in the dental information system environment are primarily due to data security risks, lack of data interoperability, and fragmented legal governance (Schleyer et al., 2011; Schwendicke et al., 2025). Therefore, it is essential for the dental health information technology infrastructure to be aligned with the HIPAA Privacy and Security rules to protect patient data and build trust. Future efforts should focus on tracking and managing the rapid growth of technological progress

and changing environments to ensure compliance with all HIPAA requirements and continue success with governance improvements. The rise in integration of artificial intelligence, cloud computing and tele dental systems into dental IT means that HIPAA compliance will require ongoing policy changes and will continue to require training and enforcement in order to ensure that policy changes do not lag behind technological advances (Schwendicke et al., 2025). For HIPAA to provide adequate protection for dental health information, especially in small practices, there needs to be improved support for small practices and improved enforcement to adjust to rapidly changing technologies.

Bibliography

Health Insurance Portability and Accountability Act of 1996, 42 U.S.C. § 1320d et seq.

Health Information Technology for Economic and Clinical Health Act, Pub. L. No. 111-5, 123 Stat. 226 (2009).

Standards for Privacy of Individually Identifiable Health Information, 45 C.F.R. §§ 164.500–164.534.

Security Standards for the Protection of Electronic Protected Health Information, 45 C.F.R. §§ 164.302–164.318.

Breach Notification for Unsecured Protected Health Information, 45 C.F.R. §§ 164.400–164.414.

HIPAA Administrative Requirements and Enforcement, 45 C.F.R. §§ 160.300–160.316.

Uses and Disclosures of Protected Health Information, 45 C.F.R. § 164.502.

Access of Individuals to Protected Health Information, 45 C.F.R. § 164.524.

Chasteen, J. E., Murphy, G., Forrey, A., & Heid, D. (2003). The Health Insurance Portability and Accountability Act and the practice of dentistry in the United States: Electronic transactions. *Journal of Contemporary Dental Practice*, 4(4), 108–120.

Edemekong, P. F., Annamaraju, P., Afzal, M., et al. (2024). Health Insurance Portability and Accountability Act (HIPAA) compliance. <https://www.ncbi.nlm.nih.gov/books/NBK500019/>

Estai, M., Kanagasingam, Y., Xiao, D., Vignarajan, J., Bunt, S., Kruger, E., & Tennant, M. (2018). A systematic review of the research evidence for the benefits of tele dentistry. *Journal of Telemedicine and Telecare*, 24(3), 147–156. <https://doi.org/10.1177/1357633X16689433>

Fernandez-Aleman, J. L., Senor, I. C., Lozoya, P. A., & Toval, A. (2013). Security and privacy in electronic health records: A systematic literature review. *Journal of Biomedical Informatics*, 46, 541–562. <https://doi.org/10.1016/j.jbi.2012.12.003>

Gupta, S., Kapoor, M., & Debnath, S. K. (2025). Cybersecurity risks and threats in healthcare. In *Artificial intelligence enabled security for healthcare systems*. https://doi.org/10.1007/978-3-031-82810-2_3

Jack, W. (2024). Cybersecurity threats in healthcare IT: Risk mitigation strategies for enhanced data privacy and resilience in Industry 4.0. <https://doi.org/10.13140/RG.2.2.25455.14244>

Jayatissa, P., & Hewapathirane, R. (2023). A review of dental informatics: Current trends and future directions. *ArXiv*. <https://doi.org/10.48550/arXiv.2307.03686>

Henry, K. (2025, January 12). HIPAA for dental practices: Complete guide. *Accountable*. <https://www.accountablehq.com/post/hipaa-compliance-for-dental-offices>

National Institute of Standards and Technology. (2024). The NIST cybersecurity framework (CSF) 2.0. <https://www.nist.gov/cyberframework>

Pai, S. S., & Zimmerman, J. L. (2002). Health Insurance Portability and Accountability Act (HIPAA): Implications for dental practice. *Dental Today*, 21(10), 106–111

Rittenhouse, D. R., et al. (2017). Increased health information technology adoption and use among small primary care

physician practices over time: A national cohort study. *The Annals of Family Medicine*, 15(1), 56–62. <https://doi.org/10.1370/afm.1992>

Scarano A, Inchingolo F, Dipalma G, Mastrangelo F. A Digital Clinical Records Versus Paper Records in Dental Practice: A Comparative Study. *Biomed Res Int*. 2025 Dec 28; 2025:5527391. doi: 10.1155/bmri/5527391. PMID: 41473735; PMCID: PMC12745653.

Schleyer, T. K. L., Thyvalikakath, T. P., Spallek, H., Torres-Urquidy, M. H., Hernandez, P., & Yuhaniak, J. (2006). Clinical computing in general dentistry. *Journal of the American Medical Informatics Association*, 13, 344–352. <https://doi.org/10.1197/jamiaM1990>.

Schwendicke, F., Samek, W., & Krois, J. (2020). Artificial intelligence in dentistry: Chances and challenges. *Journal of Dental Research*, 99(7), 769–774. <https://doi.org/10.1177/0022034520915714>.

Simon, S. R., Keohane, C. A., Amato, M., Coffey, M., Cadet, B., Zimlichman, E., & Bates, D. W. (2019). Lessons learned from electronic health record implementation in dental and medical practices. *Journal of the American Medical Informatics Association*, 26(11), 123–130. <https://doi.org/10.1093/jamia/ocz056>.

U.S. Department of Health & Human Services. (n.d.). Health Insurance Portability and Accountability Act of 1996 (HIPAA). <https://www.hhs.gov/hipaa>.

U.S. Department of Health & Human Services, Office for Civil Rights. (n.d.). HIPAA breach portal. <https://ocrportal.hhs.gov>.

U.S. Department of Health & Human Services, Office for Civil Rights. (2024). HIPAA and 42 CFR Part 2 guidance. <https://www.hhs.gov/hipaa>.

Zakaria, S. (2024). Data security and privacy in dental informatics. *Journal of Health & Medical Informatics*, 15(5). <https://doi.org/10.37421/2157-7420.2024.15.548>.